THE SUPPORT OF A SHEAF OF RINGS ON THE SMALL ETALE SITE IS CLOSED

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument. It is short because its content is the reduction of the vanishing of a ring stalk to the vanishing of the germ of the unit section under the chapter's own convention that a ring is 0 if and only if $1 = 0$. No step is omitted and no closing sentence is a placeholder.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217). The class block and the externaldocument lines of the native preamble are not included, because the assembler supplies the wrapper class and those external documents are not part of this material.

Allowed standard prerequisites (editorial).

- Support of a sheaf of rings on a site. Same chapter, the definition of the support of a sheaf of rings on S_etale as the set of points where the stalk is nonzero. Supplies the object whose closedness is asserted; the counted proof identifies it with the support of the unit section.
- The unit section of a sheaf of rings and the conventions on the zero ring. Same chapter, the structure of a sheaf of rings on S_etale, with the convention that a ring is zero exactly when $1 = 0$. Supplies the identification of the support of the sheaf of rings with the support of its unit section.

Lemma 0.1. *The support of a sheaf of rings on $S_{étale}$ is closed.*

Proof. This is true because (according to our conventions) a ring is 0 if and only if $1 = 0$, and hence the support of a sheaf of rings is the support of the unit section. □

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